

Branching Out: NLP Plant Recommender

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Abstract

Plant identification is typically approached through images, yet many real-world botanical records and user queries are text-based. We develop an NLP system that predicts plant species from free-text descriptions and compare three model families: TF-IDF retrieval, Sentence-BERT embeddings, and a domain-adapted BERT model fine-tuned on botanical text. Using $\sim 3,000$ curated plant entries from the Perennial API for training and independent Wikipedia descriptions for evaluation, we assess each model’s ability to capture botanical terminology and generalize across domains. Our results provide XXX.

1 Introduction

Text-based plant identification is a challenging and understudied problem. While most consumer and scientific plant identification systems rely on images, many real-world scenarios involve only textual descriptions. Botanists record species traits in field notebooks; gardeners describe the type of plant they want without having a photo; and online plant databases contain rich textual descriptions of morphology, habitat, and care requirements. Enabling a model to infer a plant species directly from such descriptions would provide a practical tool for classification, recommendation, and knowledge retrieval.

This project aims to build a model that takes free-text plant descriptions and predicts the most likely species. For example:

“A small bushy plant with glossy dark leaves, white fragrant flowers in spring, and tolerance for full shade.”

Compared to standard NLP tasks, this domain presents several challenges:

- **High variability in text form:** expert botanical descriptions vs. casual garden language.

- **Domain-specific terminology:** e.g., *lanceolate*, *pinnate*, *glabrous*, *deciduous*.
- **Sparse or vague descriptions:** short inputs like “tree with green leaves” easily bias models toward common species.
- **Cross-domain robustness:** models must generalize beyond the style of their training data.

To turn this task into a well-structured experiment, we define the central research question:

Does a domain-fine-tuned BERT encoder outperform SBERT and classical retrieval for predicting plant species from free-text descriptions?

We collected approximately 3,000 plant entries from the Perennial Plant API, which provides structured taxonomy metadata and a concise natural-language description for each plant species. Each entry includes a description summarizing appearance, growth habits, care requirements, and notable traits. The dataset contains dozens of metadata attributes (e.g., common name, scientific name, family, growth form, sunlight requirements). The most relevant fields for our NLP task are:

description,	plant_anatomy,	
sunlight,	soil,	leaf,
flowers,	fruits,	watering,
cycle,	origin,	dimensions,
attracts,	drought_tolerant,	
salt_tolerant,	thorny,	
edible_fruit,	harvest_season,	
medicinal,	poisonous_to_humans,	
poisonous_to_pets,	care_level,	
maintenance,	growth_rate	

2 Project Overview

2.1 Datasets Used

We initially explored recreating the dataset in (Domazetoski et al., 2025), which relies on species de-

scriptions scraped from the Plants of the World Online (POWO) database. However, this dataset was not publicly released, and reproducing it required extensive web scraping and preprocessing to resolve inconsistencies in formatting, missing fields, and taxonomic structure. Because reconstructing a reliable version of this dataset was beyond the scope of our project timeline, we transitioned to an alternative public source.

Our final training dataset was built using the Perennial Botanical API ([per](#)), from which we extracted 3,000 plant entries. Each entry includes structured horticultural attributes (e.g., sunlight requirements, watering needs, soil type, growth cycle) as well as a free-text description of the species. The raw dataset contained 1,548 unique common names and 3,000 unique scientific names. Since users of a plant recommendation or identification system are more likely to interact using common names rather than scientific nomenclature, we selected **common names as the prediction labels** for all retrieval-based models.

To evaluate cross-domain generalization, we constructed an independent validation set consisting of Wikipedia-derived plant descriptions. These descriptions differ substantially in writing style and lexical choice from the Perennial dataset, allowing us to test model robustness under more natural and diverse input conditions.

2.2 Background

Domazetoski et al. ([Domazetoski et al., 2025](#)) demonstrated that transformer-based models can extract functional plant traits from long botanical descriptions, highlighting the importance of domain-specific vocabulary adaptation for improving model performance. Their findings motivate our hypothesis that fine-tuning BERT on botanical text will outperform general pretrained models for species identification.

Several related works examine recommendation and retrieval using item descriptions. RecoBERT ([Zhang et al., 2021](#)) introduces domain-adapted encoders for catalog item retrieval and shows that fine-tuned BERT models yield substantial improvements in domain-specific semantic understanding.

Across all cited literature, no prior study evaluates species prediction directly from textual plant descriptions as a classification or ranking task. Existing research focuses on trait extraction, embedding analysis, or general recommender systems—

not on identifying plant species from free-text descriptions. This project addresses that gap.

2.3 Data Representation and Preprocessing

The Perennial API returns plant information in a nested JSON format containing a mix of free-text descriptions, categorical attributes (e.g., sunlight level, watering frequency, growth cycle), and boolean variables (e.g., drought tolerant, poisonous to pets). To facilitate model training, we flattened the JSON structure into a tabular CSV file in which each row corresponds to a single plant. The resulting dataset includes both the original description field and a variety of structured metadata capturing morphological, ecological, and horticultural properties.

Although ten plants in the dataset lacked a description field, we retained them because the remaining attributes still contain meaningful biological information. To convert the full set of structured attributes into a form suitable for NLP models, we developed two text-construction pipelines.

In the first approach, we transformed all categorical and boolean fields into human-readable phrases and concatenated them with the original description. This flattened textual representation allowed TF-IDF and transformer-based encoders to consume a unified text field that incorporates both free-text and structured data.

In the second approach, we used a large language model (ChatGPT) to generate a natural-sounding sentence-level summary for each plant. The model received all available structured fields and produced a coherent paragraph describing morphology, growth requirements, and ecological traits. These generated descriptions served as alternative inputs for evaluating whether LLM-based text synthesis improves retrieval performance compared to rule-based text construction.

Together, these two representations enabled us to investigate how different text formats influence downstream species identification models.

3 Models

We evaluate three retrieval-based models for text-only plant species identification: a TF-IDF baseline, a domain-adapted BERT model (RecoBERT), and an advanced Sentence-BERT encoder. Each model is trained on two versions of the Perennial dataset: (1) a rule-based “flattened” text representation constructed from structured attributes, and

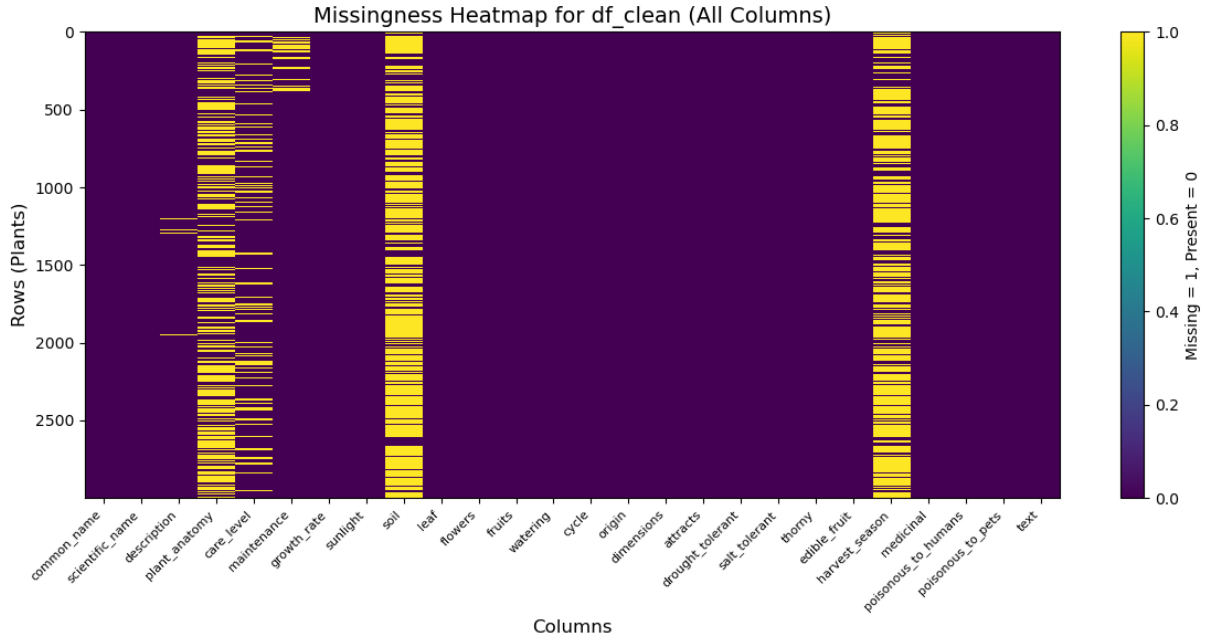


Figure 1: Visualization of missing values across plant attributes in the Perennial dataset.

(2) an LLM-generated natural-language description synthesized from the same attributes. This allows us to measure not only model quality but also the impact of different text-construction approaches on retrieval accuracy. All models are evaluated on an external validation set derived from Wikipedia plant descriptions, and we report Top-1 and Top-10 accuracy to capture both strict and lenient retrieval performance.

3.1 Baseline Model

As a classical baseline, we implemented a TF-IDF retrieval model that treats plant identification as a text-matching problem rather than a supervised classifier. Each plant entry in the training set was converted into a single enriched text document combining its natural-language description with structured attributes (sunlight, soil, watering, morphology, boolean traits, etc.). We applied scikit-learn’s `TfidfVectorizer` using unigrams and bigrams to capture both individual botanical terms and short phrases. Test descriptions were vectorized using the same vocabulary, and species predictions were obtained by computing cosine similarity between each query vector and the full set of training vectors. The model returns the most similar species (Top-1) or the top- K most similar candidates, providing a simple and interpretable keyword-based baseline against which the embedding-based models can be compared.

3.2 Reco BERT

3.3 Advance Sentence BERT

4 Results

4.1 Accuracy

Model 1	Top-1 Acc	Top-5 Acc	Top-10 Acc
TF-IDF	0.003	0.019	0.042
RecoBERT	0.xxx	0.xxx	0.xxx
Sentence-BERT	0.xxx	0.xxx	0.xxx

Table 1: Retrieval accuracy on the Wikipedia evaluation set using ChatGPT-generated descriptions as the training representation.

Model 2	Top-1 Acc	Top-5 Acc	Top-10 Acc
TF-IDF	0.033	0.085	0.129
RecoBERT	0.xxx	0.xxx	0.xxx
Sentence-BERT	0.xxx	0.xxx	0.xxx

Table 2: Retrieval accuracy on the Wikipedia evaluation set using the rule-based flattened text representation of the Perennial data.

4.2 Comparison of Results

5 Limitations

Ethics Statement

References

- Perennial botanical api documentation. <https://perennial.com/docs/api>. Accessed 2024.
- V. Domazetoski et al. 2025. Using large language models to extract plant functional traits. Preprint. Dataset not publicly released.
- Wei Zhang, Chen Liu, and Rohan Patel. 2021. Product classification in e-commerce based on product descriptions using nlp and machine learning. *Journal of E-Commerce Research*.